

Doctoral Preliminary Exam for Yifan Yang

Written Assignment 1

From: Lei Zou

1. Fundamental GIScience Knowledge

- 1.1 What is geospatial AI? How is geospatial AI distinguished from traditional machine learning and AI methods applied to geospatial datasets? What are the main research themes/trends in geospatial AI?

GeoAI is a concept that is hard to pin down. After asking many scholars in the GIS field, my conclusion is that it is an inherently interdisciplinary and complex idea, one that can only be answered by bringing together Geographic Information Science, Artificial Intelligence, and geographic data science. **In one sentence, GeoAI, or geospatial artificial intelligence, is a concept, or a research field,** that seeks to apply AI methods at the intersection of geospatial data, geographic problems, and spatial decision-making.

In fact, the question of how GeoAI fundamentally differs from simply applying traditional machine learning to spatial data is itself the best way to answer what GeoAI is. We can again think about this through three levels: the model, the data, and the computation.

At the model, or algorithmic, level, ordinary machine learning typically assumes that samples are independent and identically distributed. GeoAI, by contrast, must account for spatial autocorrelation, spatial heterogeneity, scale effects, the Modifiable Areal Unit Problem (MAUP), boundary effects, and regional transfer.

At the data level, the focus is on geospatial data and on how models are validated against it. Conventional validation often splits training and test sets at random, whereas GeoAI places far greater emphasis on spatial cross-validation and cross-region generalization, as illustrated by the training strategy in Hao's study presented yesterday.

At the computational level, traditional machine learning usually works with data of fixed dimension and clear boundaries, so data loading and batching are relatively standardized, and the underlying operators are highly optimized as standard matrix multiplication, which GPUs accelerate extremely well. Geospatial data, by contrast, is multi-scale, multimodal, and high-dimensional (spatiotemporal, or 4D). Geocomputation also involves a great deal of complex topological reasoning (such as intersection and containment), non-Euclidean geometry (such as spherical distance and geodesics), and spatial joins. This is also why CyberGIS, the high-performance and distributed computing infrastructure for geospatial analysis, is such an important concept here.

The main research themes and trends in GeoAI are, to a large extent, driven by the rapid development of AI itself. This is something that we, as GIScientists, have to acknowledge and reflect on. As AI gives rise to new topics, we can take each of them and apply, study, and rethink it in the context of the geospatial problems we work on.

Based on what I presented in an earlier group meeting, I see four major directions, all of which are connected to my own research.

1. ***From task-specific models toward Geospatial Foundation Models (GFMs)***. This direction is very active right now, but it depends heavily on computing power and data. Our own work focuses more on benchmarking and critical reflection than on building models at the architectural level.

2. ***Autonomous GeoAI and Multi-Agent Systems***. This is exactly the aim and direction of my Case 3, even though the design of Case 3 is comparatively simple.

3. ***Trustworthy GeoAI and Spatially-Constrained Learning***. This is a direction the academic community should spend far more time on: how to use GeoAI more responsibly and more effectively.

4. ***Earth Observation for Disaster Resilience and Digital Twins***. **As models and data continue to grow in scale, our capacity to take on tasks becomes stronger. This means people are no longer satisfied with simply getting an answer to a question. They increasingly expect more interactive experiences, and richer, more diverse forms of data, through which to understand disasters and, more broadly, the very concept and practice of geographic observation.**

1.2 What is generative AI, and why is it important in geography/GIScience? Explain the relevance and distinctions between generative AI and concepts like agentic AI and Foundation AI.

Before defining what Generative AI is, I think it is important to establish a broader distinction: AI can generally be divided into two families, **Discriminative AI and Generative AI**.

Discriminative AI is about recognition, classification, prediction, and pattern finding. It analyzes existing data, extracts features, and makes judgments about new inputs. This is also the methodological tradition that Geographic Information Science has long relied on. My Case 1, which performs disaster damage classification, is a clear example of discriminative AI.

Discriminative AI, however, has no ability to generate or to explain. This is precisely where Generative AI becomes important. Its core purpose is to generate new content, or more precisely,

new data. Unlike discriminative AI, which learns how to carry out a task such as disaster classification, **Generative AI learns what the "essential characteristics" of a disaster actually are, and then transforms that understanding into other forms of data.**

These forms include text that is easier for people to understand (my Case 2), disaster reports that combine imagery and text (Case 3), the generation of one type of image from another, such as satellite imagery to street view (Case 4), and even the generation of 3D data (my planned Case 6). A familiar everyday example is Google Maps: with a single switch, you can move from the ordinary map to a novel 3D view and take a virtual walk through a neighborhood. An experience like this requires capturing large real-world areas from the air, the idea behind a project proposed by Eustace at Google.

I will include this part in the background section of my preliminary exam.

The fundamental reason Generative AI matters for Geography and GIScience is that it attempts to "understand" what geography, and GIScience as a discipline, actually are. **The key to this lies in the scale of the model, the volume of data it is trained on, and the computing power required behind it.** It is worth pausing to compare the typical model size of Discriminative AI with that of Generative AI, because the difference in scale is substantial.

Generative AI matters because it tries to answer a far wider range of geographic and GIScience questions. Just as importantly, its very existence raises many new questions that are themselves worth researching and discussing, and that, in itself, is valuable for the academic community.

GIScience does not only need to "predict a label." It also needs to turn complex geographic data into interpretable spatial knowledge, into decision scenarios, and into representations that people can understand. At the same time, Generative AI is prone to hallucination and often disregards spatial constraints. Confronting these weaknesses is precisely where geography can give back to, and contribute to, Generative AI.

Agentic AI and Foundation Models are both relatively new terms and concepts. I find it useful to think about all of them through three layers: the scale of the training data, the complexity of the model itself, and the computing power that supports them.

Agentic AI emphasizes a system's operating mechanism and its autonomy. It typically uses Generative AI, and large language models in particular, as its "cognitive brain." The generative model is responsible for understanding the goal and producing a plan, while the agent is responsible for breaking that plan down into a sequence of concrete, multi-step actions.

Foundation Models emphasize large-scale pretraining. Supported by substantial computing power, the goal is to train a single model with strong general-purpose capabilities that can be transferred and broadly fine-tuned across many downstream tasks. Importantly, foundation models are not necessarily generative.

My Case 3 illustrates how these three work together. Foundation models provide the underlying vision-language ability to process multimodal geographic data, such as street-view and remote sensing imagery. Generative AI then gives the system its capacity for logical reasoning, commonsense question answering, and the generation of human-readable reports. Finally, the Agentic AI built on top of these layers is what delivers a high degree of autonomy.

1.3 What are the emerging challenges in disaster resilience? What are the common research topics in understanding and enhancing disaster resilience through geospatial approaches?

My understanding is that disaster resilience has always been an old problem. What has changed is that, as models and data become more powerful and more abundant, we are now able to revisit these long-standing questions and, at the same time, raise new ones. The challenges I would summarize are as follows.

1. Disaster resilience is a complex process, and a complex system. My sense is that we have traditionally taken an ontological approach, abstracting data out of the disaster-resilience process in order to interpret and answer questions about it. In reality, however, compound and cascading disasters are becoming increasingly frequent, and a single model trained on a single data source cannot capture what is fundamentally a process-level problem.

2. Spatial Inequality and Hidden Vulnerability. This is my most important reflection. There is almost a Matthew effect at work here: disadvantaged groups are often exposed to higher disaster risk, are underrepresented in the digital space, and are poorly served by existing insurance systems. How to precisely measure, and ultimately reduce, these micro-scale forms of social vulnerability is a central pain point of resilience building.

3. Whether disaster-resilience models can generalize. A disaster produces an enormous volume of multimodal data, including remote sensing, street view, and social media. Yet in emergency response, where every second counts, there is rarely enough time to collect and label data for this particular "unseen event" and to fine-tune a model accordingly.

Common research topics that rely on spatial methods include, but are not limited to, disaster hazard mapping, exposure and vulnerability assessment, social vulnerability indices, the simulation of floods, wildfires, landslides, and storm surge, remote sensing damage detection, shelter and evacuation accessibility, the service areas of critical facilities, road disruption and rescue routing, population mobility, social media sensing, recovery monitoring, the construction of resilience indicators, scenario simulation, and risk communication. **Building on my current research interests, I am particularly drawn to next-generation 4D disaster digital twins, multi-agent scenario simulation, and interactive disaster decision-making.**

1.4 What are the advantages and challenges of incorporating geospatial big data and AI techniques in disaster resilience research and practical applications?

As I have said throughout, the greatest advantage lies in combining spatial big data with AI, which is fundamentally reshaping the paradigm of disaster management, moving it from a passive model of "post-disaster response" toward proactive "intelligent cognition." This is also what makes the research directions we are now pursuing possible.

1. Holistic Perception via Multi-Modal Data. By combining macro-scale satellite remote sensing (overhead imagery), micro-scale street-level imagery, and dynamic social sensing data such as VGI and social media trajectories, AI can capture a fine-grained and complete picture of a disaster that no single traditional data source can provide. (For example, my Case 1 and Case 4.)

2. Near Real-Time Automation and Zero-Shot Generalization. Given how suddenly disasters strike, traditional, fully manual loss assessment is extremely time-consuming. GeoAI, and agentic pipelines in particular, can deliver out-of-the-box, zero-shot damage assessment, greatly improving the time efficiency of emergency response. (For example, my Case 3.)

3. Autonomous GeoAI for Science. We can use AI to pose new and broader disaster-resilience research questions. We are also already using AI tools to support the programming, the figure-making, and even the writing involved in our research.

The challenge is that, precisely because we are working with geospatial big data and AI, we may arrive at mistaken understandings and decisions, and these errors are often difficult to notice.

Black-box Nature and Spatial Hallucinations. Purely data-driven AI often lacks any grasp of physical laws or of the fundamental laws of geography, such as topological connectivity, which makes it prone to "spatial hallucinations" that violate physical reality. In disaster decisions, where lives and property are at stake, the lack of interpretability and trustworthiness is the greatest barrier to real-world deployment. *In my view, this is not fundamentally a problem of AI or of data. It is a problem of understanding the question incompletely, and of making subjective errors when choosing the data. In the same spirit, there is no single "correct" model, only a model that is appropriate for the problem at hand.*

Data Representativeness Bias and Spatial Inequality. Spatial big data, especially crowdsourced and social media data, often carries serious sampling bias. Low-income neighborhoods and elderly communities with weak digital infrastructure tend to be "invisible" in the data. If an AI model relies on such data uncritically, it can widen the "digital divide" in emergency resource allocation and deepen spatial inequality.

2. Ph.D. and Career Planning

2.1 Dissertation Overview:

Please provide an overview of your Ph.D. dissertation's structure and discuss its intellectual merits and broader impacts. Include details such as the number of chapters, their content, and the potential challenges your research will address. Describe any innovative datasets, products, or novel knowledge your research will generate. Explain how your dissertation will address existing gaps in GIScience research and outline the potential benefits for researchers, governments, and the public.

Title: **Cross-View Geospatial Artificial Intelligence for Disaster Damage Assessment: From Ground-Level Perception to Immersive Decision Support.**

One-line thesis. This dissertation builds cross-view geospatial artificial intelligence that lets us see, describe, and govern disasters, advancing from hyperlocal ground-level perception, through reliable and autonomous cross-view reasoning, to immersive, human-centered decision support.

This dissertation adopts a manuscript-style structure of nine chapters: an introduction, a literature review, six case-study chapters (each corresponding to one paper), and a concluding synthesis. The six case chapters are organized into three thematic groups (A, B, C) that follow the central logic of See, Describe, and Govern.

- Chapter 1, Introduction. Establishes the research motivation, the two complementary ways of observing a disaster (overhead and ground), the three current challenges, the three-layer GeoAI framework (Observation, Intelligence, Delivery), and the three objectives that organize the work.

- Chapter 2, Literature Review.

Group A — Discriminative AI for Disaster Assessment (See). This first stage concerns the contribution of discriminative AI to geospatial disaster assessment. It asks how we can make AI see geospatial data better and more responsibly, and how we can design its algorithms more responsibly. It treats damage assessment as a problem of recognition and classification, while insisting that reliability, interpretability, and responsible labeling are prerequisites rather than afterthoughts.

- Chapter 3 (Case 1), Hyperlocal Bi-Temporal Street-View Assessment (published, CEUS 2025). Shows that using pre-disaster street-view as a no-damage baseline in a dual-channel network raises hyperlocal assessment accuracy from 66.14% to 77.11%.

- Chapter 4 (Case 2), DamageArbiter. A CLIP-enhanced, disagreement-driven arbitration framework that reaches 75.85% accuracy and 0.619 MCC using only label-free, inference-time features, while mitigating model overconfidence.

Group B — Multimodal Generative AI and Agentic Cross-View Reasoning (Describe). Here the work moves beyond asking AI merely to see a disaster. It asks AI to understand damage assessment in a more generalizable way and to describe it through outputs that people can read and act on, including structured reports, synthesized cross-view imagery, and honest evaluations of when cross-view evidence actually helps. This stage brings together multimodal generative AI, autonomous agents, and cross-view synthesis and fusion.

- Chapter 5 (Case 3), RAPID. A reproducible multi-agent workflow (perception, restoration, recognition, reasoning) for zero-shot, interpretable cross-view assessment; it reaches 0.92 multi-disaster classification accuracy and automatically generates location-specific response reports.
- Chapter 6 (Case 4), Satellite-to-Street Synthesis. Generative cross-view synthesis evaluated with a Structure-Aware Evaluation Framework, revealing a trade-off between visual realism and structural fidelity in disaster scenes.
- Chapter 7 (Case 5), Conflict-Aware Evaluation (CAE). A reporting protocol showing that the value of cross-view fusion depends on the spatial observation regime, and that spatial leakage can create a false impression that "cross-view fusion helps."

Group C — Immersive Delivery and the Governance of Disasters (Govern). After seeing and describing, this final stage asks how AI can give people better feedback and help govern disasters, and what the future of interaction among AI, people, and geographic information should look like.

Chapter 8 (Case 6), Cross-View Splatter (planned flagship). Feed-forward synthesis of interactive, georeferenced 3D disaster scenes, representing the immersive "last mile" of decision delivery.

Chapter 9, Conclusion. Synthesizes the contributions, limitations, and future directions of the work, and revisits the See, Describe, and Govern progression that runs through the dissertation.

At the data level, this research contributes a number of datasets that integrate pre- and post-disaster street view, satellite imagery, multi-hazard samples, and paired cross-view spatial evidence. At the method level, I propose a dual-channel street-view assessment network, DamageArbiter, the RAPID multi-agent workflow, a Structure-Aware Evaluation Framework, a Conflict-Aware Evaluation protocol, and the planned Cross-View Splatter. Each of these is a novel method and a concrete contribution to GeoAI. At the knowledge level, the dissertation answers when ground-level evidence is useful, when cross-view fusion is likely to fail, how generative disaster scenes should be evaluated, and how AI can move from "recognizing damage" toward "explaining disasters" and "supporting governance."

Addressing Existing Gaps in GIScience

My dissertation is positioned at the deep intersection of Geographic Information Science (GIScience), Artificial Intelligence, and disaster resilience. This interdisciplinary vantage point allows the work to close four core gaps that conventional spatial machine learning has long left open.

1. An epistemological leap: from "predicting a label" to "generating trustworthy spatial knowledge."

Traditional spatial machine learning (discriminative AI) often stops as soon as it has "predicted a label," such as deciding whether a building is damaged. This black-box classification output is difficult to act on in the complexity of post-disaster response. My research fills this gap by pushing GeoAI beyond mere "perception output" toward the generation of interpretable, trustworthy, and actionable spatial knowledge. By introducing multi-agent reasoning (as in Case 3, RAPID), the system can understand a disaster situation and produce structured, natural-language decision recommendations, marking a qualitative step up in spatial intelligence.

2. Bridging the macro and micro divide: hyperlocal cross-view fusion.

Existing disaster remote sensing assessment relies heavily on a top-down, overhead perspective, which leaves serious physical blind spots (for example, it cannot detect a flooded ground floor or facade-level structural damage). By introducing and aligning bi-temporal, hyperlocal street-view data (Case 1), this dissertation bridges the gap between macro-scale Earth observation and micro-scale ground truth, achieving deep, multi-scale, cross-view feature fusion.

3. Establishing an evaluation standard for Trustworthy (Fiduciary) GeoAI.

In current GeoAI research, model reliability, interpretability, and uncertainty are too often treated as afterthoughts, and many cross-view fusion studies have even fallen into a false sense of progress caused by spatial leakage. My research fills this methodological gap by treating these concerns as primary objectives from the very start of the design: the disagreement-driven arbitration mechanism that mitigates overconfidence (Case 2, DamageArbiter) and the Conflict-Aware Evaluation protocol (Case 5, CAE) together establish a rigorous trustworthiness standard for geospatial foundation models.

4. Crossing the "last mile": from static mapping to immersive delivery.

A major gap facing GIScience today is the disconnect between back-end AI perception outputs (spatial tensors) and front-end, human-centered decision-making. By exploring 3D generative techniques such as Cross-View Splatter (Case 6), this dissertation turns complex damage-assessment results into georeferenced, interactive, immersive 3D disaster scenes. This fills the long-missing "delivery" gap in disaster informatics, transforming traditional one-way, passive disaster mapping into two-way, immersive decision support.

1. For Researchers: Advancing a "Responsible GeoAI" Research Philosophy

My research does more than provide new tools, models, or methods. It also aims to lead and to communicate a research philosophy and a new perspective, that of a responsible and rigorous GeoAI. Building on Case 4 (generative cross-view synthesis with a Structure-Aware Evaluation framework) and Case 5 (Conflict-Aware Evaluation, CAE), I hope my dissertation offers the broader GIScience community a benchmark protocol and an open-source workflow for assessing the true value of cross-view fusion. By empirically revealing the pitfalls that arise under different spatial observation regimes, it directly corrects a false sense of progress in the field, namely the assumption that "fusion always helps," which often stems from spatial leakage.

2. For Governments: Delivering an Out-of-the-Box, High-Trust Decision Pipeline

Building on Case 3 (the RAPID multi-agent workflow) and Case 2 (the DamageArbiter arbitration mechanism), this research delivers a genuinely out-of-the-box, zero-shot, automated damage diagnosis for emergency-management agencies such as FEMA. It not only greatly compresses post-disaster response time, but also moves beyond traditional black-box labels: it can automatically produce structured, location-specific natural-language response reports that include explicit confidence boundaries to mitigate model overconfidence, enabling safe and rapid damage assessment with minimal human intervention. If my research has value, then seeing it genuinely put into practice is my deepest hope.

3. For the Public: Human-Centered Immersive Delivery and Spatial Equity

This research seeks to close the "last mile" through which technology reaches the public. Through Case 6 (Cross-View Splatter, interactive 3D scenes), it turns complex and opaque AI spatial-tensor outputs into georeferenced, immersive 3D disaster scenes for immersive decision support. This intuitive human-computer interface lets ordinary community residents, even those with no technical background, directly understand the damage to their own homes, making community-level, micro-scale mitigation decisions more transparent, more democratic, and more accessible. Building on this foundation, I hope to develop it into a product, and perhaps even to launch a company.

2.2 Publication Plan:

Share your plan for publications during your Ph.D. program. List the titles, brief descriptions, current progress status, and the target journals for each planned publication.

Group A — Discriminative AI for Disaster Assessment (See)

#: 1 Title: Hyperlocal Disaster Damage Assessment Using Bi-Temporal Street-View Imagery and Pre-Trained Vision Models

Brief description: Uses pre-disaster street view as a no-damage baseline in a dual-channel network; raises hyperlocal accuracy 66.14% → 77.11% (Hurricane Milton).

Status: Published (2025)

Target venue: Computers, Environment and Urban Systems (CEUS) 

#: 2 Title: DamageArbiter: A CLIP-Enhanced Multimodal Arbitration Framework for Hurricane Damage Assessment from Street-View Imagery

Brief description: Disagreement-driven, label-free arbitration of ViT and CLIP; 75.85% acc / 0.619 MCC while mitigating overconfidence.

Status: ~95%, ready for submission

Target venue: International Journal of Applied Earth Observation and Geoinformation (JAG); alt: International Journal of Disaster Risk Reduction.

Group B — Multimodal Generative AI & Agentic Cross-View Reasoning (Describe)

#: 3 Title: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

Brief description: Four-agent, zero-shot, fine-tuning-free workflow; 0.92 multi-disaster classification; auto-generates response reports.

Status: ~90%, under review

Target venue: ACM SIGSPATIAL '26 (Applications)

#: 4 Title: Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models

Brief description: VLM-guided + Disaster-MoE synthesis with a Structure-Aware Evaluation Framework, reveals the realism vs. structural-fidelity trade-off.

Status: Accepted (venue to confirm)

Target venue: Accepted to IGARSS 2026 — IEEE International Geoscience and Remote Sensing Symposium. 

#: 5 Title: When Cross-View Helps Most: Conflict-Aware Disaster Triage Across Spatial Observation Regimes

Brief description: Conflict-Aware Evaluation (CAE) protocol; shows fusion value tracks the observation regime, and that spatial leakage can fabricate "cross-view helps."

Status: ~50%, under review

Target venue: ACM SIGSPATIAL '26 (Experiments) → extension to Transactions in GIS or International Journal of Disaster Risk Reduction

Group C — Immersive Delivery & Governance (Govern)

#: 6 Title: Cross-View Splatter: Feed-Forward View Synthesis with Georeferenced Images for Disaster Damage Assessment

Brief description: Feed-forward, Gaussian-splatting-style synthesis of interactive, georeferenced 3D disaster scenes; the immersive "last mile" of decision delivery.

Status: Planned (flagship); target submission Spring 2027

Target venue: ISPRS Journal of Photogrammetry and Remote Sensing or International Journal of Geographical Information Science (IJGIS); alternative: a computer-vision venue such as the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) or the IEEE/CVF International Conference on Computer Vision (ICCV).

While a strong publication record is one of the most important parts of a PhD, my deepest aspiration is to be able to stand behind my own work and to earn the recognition of my advisor.

Beyond the six papers that form the core thesis line, I anticipate contributing to additional publications that sit outside the main argument. These include collaborative work arising from the DOE-funded project, as well as perspective and vision papers that reflect on the broader direction of GeoAI.

2.3 5-Year Career Plan:

Outline your career goals for the next five years. Specify the research areas you intend to delve into and why those areas are worth exploring. Indicate your interest in teaching or curriculum development for courses. Describe how you plan to contribute to the GIScience community through your research, education, and service activities.

I will be candid: this is the hardest part of the exam for me, and it is the question I think about most.

Trajectory. I expect to spend the next one to two years completing my PhD, because this training is foundational for everything that follows. After that, I plan to pursue postdoctoral training, which for me marks the transition into becoming an independent researcher. By roughly the fourth or fifth year, I believe I will move into industry and start a company. **That is my vision: entrepreneurship is, in a sense, the hero's dream of an ordinary person.**

Teaching and course development. I am genuinely willing to develop courses and to teach. At the same time, I want to be honest that I may not be able to prepare fully for both an academic and an industry track at once, so I expect to concentrate my energy where I can contribute the most.

Contributing to the GIScience community. There are many ways to give back, and not all of them run through a tenure-track position. For me, the most meaningful would be to build a genuinely good company: one that draws PhD students together to pursue ambitious work, and that helps address a real and pressing problem, namely the employment and career paths of GIS students.

The research area I want to go deep on, and why it is worth exploring.

The deep direction for the later stage of my PhD, and for my future company, is to treat immersive geographic experience as the "last mile" of spatiotemporal intelligence.

This idea crystallized while I was reading two seemingly unrelated, marketing-flavored pieces: Shep Hyken's Forbes article "Customer Interaction Is the Secret Weapon to Beating Your Competition," and the Google Maps Platform page on "Building Immersive Experiences." On the surface they are promotional, but read through the lens of my research philosophy, they point to a stage that researchers often overlook: how spatiotemporal intelligence actually reaches people, gets used by them, and is in turn calibrated by them.

Each can be compressed into a single line:

- Forbes (the why): interaction is not a by-product of satisfaction, it is competitiveness itself. The more you interact, the better you understand real needs and behavior, and the better the experience you can provide; personalized, continuous, omni-channel interaction compounds into relationships and retention.

- Google Maps (the how): using Photorealistic 3D Tiles, street view, aerial perspectives, and WebGL overlays, a "place" is turned from a flat basemap into a medium for storytelling, interaction, and immersion.

Connecting these to my three-layer GeoAI framework and to the stance of a "system designer," five insights are worth keeping:

1. Spatiotemporal intelligence cannot stop at the accuracy and benchmarks of the third layer; it also needs a delivery layer. However accurate a disaster-recognition or cross-view generation model is, if it stops at a metrics table it never truly reaches decision-makers or the public. Immersive geographic experience is precisely the interface that delivers the "last mile" of spatial intelligence.
2. Immersive 3D is not only display, it is an extension of perception and reasoning. Photorealistic 3D Tiles, street view, aerial views, and WebGL overlays lie on the same technical lineage as my work on cross-view imagery, street view, 3D scenes, and world models or digital twins. They can serve both as ready-made infrastructure for visualization and as a data and rendering foundation for cross-view generation and process simulation.
3. The business loop of "interact, understand the user better, serve them better" is, in essence, the real-world feedback loop of research. Replace "customer" with "decision-makers, emergency agencies, and the public," and this loop becomes the one I keep returning to: read, judge, research

and build, release an asset, collect real-world feedback, and recalibrate. Only a system that is actually used, and that keeps collecting feedback, enters a compounding cycle.

4. It follows the same governance thread as mutual visibility. Immersive experience lets people see spatial information, but the reverse question a researcher must ask is whether people can see how the system perceives, generates, and judges the place they live in. The more immersive the delivery, the more interpretability and data governance must live inside the interface, rather than being bolted on at the end.

5. For the disaster-resilience throughline, this is the natural outlet for public communication and decision support. Placing post-disaster damage, risk scenarios, and recovery progress into interactive, photorealistic 3D and street-view scenes is itself the presentation layer of disaster perception and decision support. It is what lets the path of seeing, describing, and helping govern disasters land in real public settings.

Compressed into one sentence: the value of research lies not only in making models accurate, but in delivering spatial intelligence in ways that are immersive, interactive, and open to feedback, so that the real world is both seen by the system and seen and used by people within it.

This echoes my own vision: to move beyond the boundaries of the screen and let the real world become the stage for intelligence. Immersive geographic experience hints at what that stage might look like, and at how to connect it back to the long-term, compounding loop of research, development, and real-world feedback.

3. Readiness for the Job Market

You have been provided with three job descriptions. Your task is to carefully review each job description and assess your readiness to apply for these positions. Specifically, please provide the following information:

- Express your level of interest in and suitability for each job, along with the reasons behind your interest and suitability.
- Identify the qualities, skills, and experiences that make you a strong candidate for each of these positions, highlighting your competitiveness.
- Describe your plans for personal improvement and development that will enhance your chances of being considered for these positions.

Frankly, I know these three positions very well, and I even know who was ultimately selected for each. As a result, I find myself instinctively benchmarking my own strengths and weaknesses against the profiles of those who were chosen. I think of the three as three distinct career paths: a research-support postdoc, a teaching-focused faculty role, and a research-oriented tenure-track faculty role.

For the Harvard CGA Postdoctoral Fellowship, both my interest and my fit are high. It is hard to say no to a top-tier institution, and to me the most important thing is the chance to work alongside excellent people. That is why the Harvard CGA fellowship is the position I am most interested in.

The role emphasizes spatial analysis, GIS training, research consultation, project development, and spatial AI, all of which are closely aligned with my doctoral training. As I understand it, the postdoc is also responsible for a great deal of courseware design and software development, for example the CGA and KNIME collaboration. On both counts, my PhD training and the position are a strong match.

I see my advantages as follows. First, I have a clear GeoAI research line, particularly in disaster resilience, cross-view reasoning, and multimodal geospatial intelligence. Second, I have solid skills in Python, GIS, remote sensing, deep learning, and spatial data processing, which let me support workshops, research consultation, and collaborative projects. These first two points satisfy the core requirements of the role. Third, beyond the technical fit, come soft skills and network: I already collaborate with researchers connected to Harvard CGA, such as Professor Siqin Wang, I am often invited to CGA workshops and events, and I am outgoing and active on social media, which fits the kind of visibility and community-building the Center wants to promote.

Of the three positions, the University of Iowa Tenure-Track Assistant Professor in GIScience is the one I am second most interested in, and the one that best fits my long-term academic vision. It explicitly emphasizes spatiotemporal modeling, GeoAI, geospatial foundation models, and human-environment interactions, and it looks for candidates focused on natural hazards, environmental hazards, climate resilience, and sustainability. These directions align closely with my

dissertation and my future research plans, and my work speaks directly to what the position calls for. I have always been confident in my research, and I genuinely believe in its quality.

The appeal of this role is that it would let me keep developing cross-view GeoAI and generative and agentic GeoAI for disaster damage assessment, disaster resilience, urban risk, and immersive decision support. My plan is to grow the throughline of "seeing disasters, describing disasters, and helping govern disasters" into a long-term research agenda.

This is also the position that challenges me the most. A tenure-track role expects a candidate who has already demonstrated the potential of an independent researcher: high-quality publications, a clear five-year research plan, the potential to win external funding, teaching ability, and a capacity for academic service. It is arguably the most competitive kind of position in academia. My advantage here lies in my temperament and in my conviction about my own work. Even if a figure as eminent as Dr. Goodchild were among the other candidates, I would still believe in the value of my work and present it without fear, with genuine, heartfelt passion. That fearlessness and that love for the work are, I think, my real strengths. Beyond that, I will be honest: at this stage I do not yet have the full track record the role demands.

For the Texas A&M Instructional Assistant Professor in GIST, I also have a strong passion and love for teaching, but among the three this is probably the position I am least drawn to. Realistically, however, it is likely the least competitive of the three. Even so, the prerequisite for any of this is mastering one key skill: spoken English. As a non-native speaker, I believe I need to invest as much in strengthening my spoken communication as in my research.

This position is more teaching-oriented. It asks for a strong commitment to education, the ability to teach in-person, online, and blended courses, and a focus on training geospatial technology professionals. I am well prepared to teach courses such as GIS, geospatial programming, GeoAI, remote sensing, spatial data science, and disaster GIS, and I can turn my own work on disaster assessment, street-view analysis, remote sensing imagery, and WebGIS into project-based learning.

My advantages are my enthusiasm for teaching and the teaching materials I have been preparing early on, for example my GitHub repository AutoGeoAI4Sci. By the time I graduate, that repository will already hold a rich body of teaching content.

I do not think any of us can ever be truly ready to go on the job market. That is precisely why I have set my sights on graduating sooner, so that I push myself to confront these realities rather than postpone them.

Frankly, I believe my future lies in industry, as an entrepreneur and a CEO. Rather than working for someone else, I would rather work for something of my own. Seen this way, my doctoral training becomes, in effect, training in how to become a better CEO. A CEO is responsible for three things:

finding money, finding people, and finding direction. Interestingly, these are also the core abilities a tenure-track professor needs: securing funding, building a team, and setting a research direction. The two paths share far more than they might appear to.

With that in mind, my plan for personal growth has four parts.

1. Strengthen my spoken English. If the goal is to find any kind of job in the United States, communication is the foundation of everything, and it is honestly my weakest point right now.
2. Build my resilience. We can never predict what the future holds, and many things will go wrong along the way. What matters is that, with enough practice, refinement, and honest criticism, resilience will help me accept failure with composure.
3. Deepen my scholarly ability. This is the core competitiveness of any PhD student, and of me in particular. Scholarly ability means the ability to innovate, and innovation is something I value deeply. It is also the cornerstone of the technology company I hope to build.
4. Cultivate vision and taste. Which directions to pursue, how to judge whether a direction is worth pursuing, and what taste to bring to it: different taste produces fundamentally different research, and good taste and a clear vision are exactly what a PhD should cultivate.

Finally, **I believe that all of us fear the unknown, but courage is what lets us overcome that fear. Of all the qualities I hold, courage is the one I value most.** Whatever the future brings, I trust that my TAMU family will stand beside me, face it with me, and support the decisions I make.